1) $2\,CH_3\text{-}CH_2\text{-}CH_3 \xrightarrow{773K} CH_3\text{-}CH=CH_2 + CH_2=CH_2 + H_2 + CH_4$

2) $2CH_3\text{-}CH_3 \xrightarrow{773K} CH_2=CH_2 + 2\,CH_4$

The products depends upon the nature of alkane, temperature, pressure and presence or absence of catalyst. The ease of cracking in alkanes increases with increase in molecular weight and branching in alkanes. Cracking plays an important role in petroleum industry.

6) Isomerisation:

Isomerisation is a chemical process by which a compound is transformed into any of its isomeric forms. Normal alkanes can be converted into branched alkanes in the presence of $AlCl_3$ and HCl at 298 k.

$$CH_3\text{-}CH_2\text{-}CH_2\text{-}CH_3 \xrightarrow[298\ K]{AlCl_3/HCl} CH_3\text{-}\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}\text{-}CH_3$$

This process is of great industrial importance. The quality of gasoline is improved by isomerising its components.

Uses

The exothermic nature of alkane combustion reaction explains the extensive use of alkanes as fuels. Methane present in natural gas is used in home heating. Mixture of propane and butane are known as LPG gas which is used for domestic cooking purpose. GASOLINE is a complex mixture of many hydrocarbons used as a fuel for internal-combustion engines.

Carbon black is used in the manufacture of ink, printer ink and black pigments. It is also used as fillers.

No of Carbon Atoms	State at room temperature	Major uses
1-4	Gas	Heating fuel,Cooking fuel
5-7	Low boiling liquid	Solvents, Gasoline
6-12	Liquid	Gasoline
12-24	Liquid	Jet fuel-portable stove fuel
18-50	High boiling liquid	Diesel fuel, lubricant, heating oil
50+	Solid	Petroleum jelly and paraffin wax

13.3.Alkenes:

Alkenes are unsaturated hydrocarbons that contain carbon-carbon double bond. They are represented by the general formulae C_nH_{2n} where 'n' stands for number of carbon atoms in the molecule. Alkenes are also known as olefins (in Latin - oil maker) because the first member ethene combines with chlorine gas to form an oily liquid as a product.

(I) Nomenclature of Alkenes:

Let us write the IUPAC name for the below mentioned alkanes by applying the general rules of nomenclature that we already discussed in unit No.11

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